Unit 1

EL – Analysis of Non Recursive Algorithms

1. Analyze the following algorithms and derive the time complexity

```
ALGORITHM SumEven(n)
// Input: A non-negative integer n
Sum ← 0
For i ← 1 to n do
   If i mod 2 = 0 then
      Sum ← Sum + i
Return Sum

ALGORITHM PairSum(n)
// Input: A non-negative integer n
Sum ← 0
For i ← 1 to n do
   For j ← 1 to n do
      Sum ← Sum + (i + j)
Return Sum

ALGORITHM PrefixPairAccumulation(n)
// Input: A positive integer n
Sum ← 0
For i ← 1 to n do
   Temp ← 0
   For j ← 1 to i do
      Temp ← Temp + j
   For k ← 1 to Temp do
      Sum ← Sum + 1
Return Sum
```

2. What does the following algorithm do? Analyze the algorithms and derive the time complexity.

```
ALGORITHM Func(n)
// Input: A non-negative integer n
Sum ← 0
For i ← 1 to n do
   For j ← 1 to i do
      Sum ← Sum + j
Return Sum

ALGORITHM Func2(n)
// Input: A non-negative integer n
Sum ← 0
For i ← 1 to n do
   If i mod 2 = 0 then
      For j ← 1 to n do
```

```
            Sum ← Sum + j
    Return Sum

    ALGORITHM Func3(n)
    // Input: A positive integer n
    Count ← 0
    For i ← 1 to n do
       For j ← 1 to i*i do
          Count ← Count + j
    Return Count
```

3. For each of the following pairs of functions, indicate whether the first function of each of the following pairs has a lower, same, or higher order of growth (to within a constant multiple) than the second function.
   a. $n(n + 1)$ and $2000n^2$
   b. $100n^2$ and $0.01n^3$
   c. $n^2 + n$ and n^2
   d. b. n^3 and $100n^2$
4. Prove or disprove the assertions

   a. $n(n + 1)/2 \in O(n^3)$

   b. $n(n + 1)/2 \in O(n^2)$

   c. $n(n + 1)/2 \in \Theta(n^3)$

   d. $n(n + 1)/2 \in \Omega(n)$